

Structure-guided design of peptide ligands with NPY receptor subtype selectivity

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Neuropeptide Y (NPY) regulates appetite through the closely related Y1 and Y2 receptors (Y1R and Y2R), which have opposing effects on food intake. Whereas Y1R activation promotes feeding, Y2R activation suppresses appetite, making Y2R a promising anti-obesity target. Here, we determined cryo-electron microscopy structures of the NPY–Y1R–Gi and NPY–Y2R–Gi complexes. Comparative structural analysis and mutagenesis revealed distinct subtype-specific recognition mechanisms of NPY, including the molecular basis for the selective requirement of the N-terminal residues of NPY for Y1R signaling. Guided by these structural insights, we computationally designed peptide ligands to preferentially engage Y2R while avoiding interactions required for Y1R activation. Functional characterization identified ligands that selectively activate Y2R with substantially reduced Y1R activity. Furthermore, a lead Y2R-selective ligand suppressed food intake in mice. These findings demonstrate that integration of structural biology and computational design can enable the development of subtype-selective peptide ligands for closely related GPCRs and provide a framework for precision targeting of therapeutically relevant receptor subtypes.